
Tail Probabilities for SGD Iterates with Heavy-tailed Gradient Noises

Xiaoli Li

Department of Statistics
The University of Chicago
Chicago, IL 60637, USA

Qianqian Lei

Department of Economics
The University of Chicago
Chicago, IL 60637, USA

Wei Biao Wu

Department of Statistics
The University of Chicago
Chicago, IL 60637, USA
wbwu@uchicago.edu

Abstract

We study the tail behavior of constant-stepsizes stochastic gradient descent (SGD) when the stochastic gradient noise is heavy-tailed with infinite variance. In contrast to classical Gaussian or central-limit-type analyses, our goal is to identify the exact first-order tail probability of SGD iterates, including the sharp leading constant. We develop an extreme-value-theoretic analysis based on the one-big-jump principle. First, we establish moment convergence for nonlinear SGD under centered dissipativity and weak local regularity assumptions, allowing objectives beyond the globally strongly convex and globally smooth setting. Second, for the linearized recursion, we prove an exact tail theorem under multivariate regular variation, with a constant determined by the interaction between the tail measure of the gradient noise and the stable linearized dynamics. A one-dimensional projected-tail version is then obtained as a special case along left eigendirections. Finally, we transfer the linear tail asymptotics to the nonlinear SGD iterate by showing that the nonlinear approximation error is negligible on the relevant rare-event scale. Our numerical experiments on nonlinear, threshold, and expectile recursions confirm that the proposed approximation captures empirical tail probabilities with the predicted exact constant, rather than merely providing an upper bound or a rate.

1 Introduction

Stochastic gradient descent (SGD) is a basic computational device for modern statistical learning and large-scale machine learning. In its classical form, stochastic approximation was introduced as a recursive method for solving population estimating equations [Robbins and Monro, 1951], and its averaged variants are known to attain optimal asymptotic efficiency under finite-variance conditions [Polyak and Juditsky, 1992, Kushner and Yin, 2003]. In contemporary applications, SGD is often used not only to optimize an objective, but also to produce a statistical estimator from streaming or massive data. Consequently, the distribution of the iterate itself matters and optimization-induced errors can enter confidence intervals, hypothesis tests, and other uncertainty quantification procedures. A growing literature has therefore studied asymptotic normality, covariance estimation, and bootstrap inference for SGD-type estimators [Chen et al., 2020, Fang et al., 2018, Sheshukova et al., 2025].

Most distributional analyses of SGD rely, either explicitly or implicitly, on a finite-variance picture [Zhu et al., 2023, Merad and Gaïffas, 2023]. In such settings, Gaussian approximations describe typical fluctuations around the optimum, especially after Polyak–Ruppert averaging or in a local stationary regime. When the finite-variance assumption fails, the tail probabilities that these tools are meant to describe are no longer captured by the Gaussian limit, and the resulting inference is systematically miscalculated. Such failure is not hypothetical. Empirical and theoretical studies have documented that stochastic gradients in deep learning and in heavy-tailed statistical models exhibit non-Gaussian, power-law-type behavior [Simsekli et al., 2019, Gürbüzbalaban et al., 2021]. In the

infinite-variance setting, the classical central-limit heuristic no longer describes the tail behavior of the iterate. In particular, rare deviations are no longer generated by the accumulation of many small fluctuations; they are instead driven by a *one-big-jump* event, namely a single large stochastic gradient shock that is subsequently propagated by the local dynamics.

This paper studies this rare-event regime for constant-stepsize SGD. We consider the population optimization problem

$$x^* = \arg \min_{x \in \mathbb{R}^d} M(x), \quad M(x) := \mathbb{E}_\varepsilon G(x, \varepsilon), \quad (1)$$

where $M : \mathbb{R}^d \rightarrow \mathbb{R}$ is the population objective and $G(x, \varepsilon)$ is a noise-perturbed loss. Given i.i.d. innovations ε_i with distribution Π , the constant-stepsize SGD recursion is

$$x_n = x_{n-1} - \gamma g(x_{n-1}, \varepsilon_n), \quad n \geq 1, \quad (2)$$

where g is a stochastic gradient and $\gamma > 0$ is small. Our goal is to characterize the projected tail probability of the last iterate as $\gamma \rightarrow 0$, i.e. $\mathbb{P}\{\langle \ell, x_n - x^* \rangle > u_\gamma\}$, for n large enough, where $\ell \in \mathbb{R}^d \setminus \{0\}$ specifies the linear functional of interest and u_γ is a constant that may depend on γ . The central question is not merely whether this probability is small, nor whether it admits a conservative upper bound, but whether its exact first-order asymptotic and sharp constant can be identified.

The answer is affirmative. Let $m(x) = \mathbb{E}[g(x, \varepsilon)]$, $H = \nabla m(x^*)$ and $\eta_0 = -g(x^*, \varepsilon_0)$ denote the centered innovation at the target. Under mild assumptions, we prove that for power thresholds $u_\gamma = \gamma^a$ in the rare-event regime,

$$\mathbb{P}\{\langle \ell, x_n - x^* \rangle > \gamma^a\} \sim C_\ell \frac{\mathbb{P}(|\eta_0| > \gamma^{a-1})}{\gamma}, \quad C_\ell = \int_0^\infty \mu_\eta\{z : \ell^\top e^{-sH} z > 1\} ds, \quad (3)$$

where μ_η is a non-null Radon measure (to be specified later) that characterizes the tail behavior, and the notation $\sim$ represents asymptotic equivalence. The constant has a transparent interpretation: a single extreme stochastic gradient shock, whose direction is governed by the tail measure μ_η , is transported by the stable linearized flow e^{-sH} and then observed through the functional $\ell^\top(\cdot)$. Thus (3) is an extreme-value-theoretic calibration of the tail of a nonlinear SGD iterate, not a Gaussian approximation and not merely an upper bound.

The main technical difficulty is that the one-big-jump principle is most direct for linear recursions and weighted sums, whereas SGD is nonlinear and may be nonconvex. Extreme value theory and regular variation provide precise tail tools for heavy-tailed sums, linear processes, and stochastic recursions [Bingham et al., 1987, Embrechts et al., 1997, Hult and Samorodnitsky, 2008, Mikosch and Wintenberger, 2013]. Existing heavy-tailed SGD theory has established convergence guarantees, stable-limit behavior, and high-probability bounds under weak moment assumptions or clipping schemes [Wang et al., 2021, Gorbunov et al., 2020, Cutkosky and Mehta, 2021, Nguyen et al., 2023, Blanchet et al., 2024]. In contrast, our result identifies the exact shrinking-threshold tail probability of the nonlinear constant-stepsize iterate, including the sharp constant.

1.1 Contributions

Our contributions are summarized as follows.

Moment convergence under weak local assumptions. We derive an L^r convergence theorem for constant-stepsize SGD for every $r \in (1, p]$ with $1 < p < 2$,

$$\mathbb{E}|x_n - x^*|^r \leq (1 - c_r \gamma)^n |x_0 - x^*|^r + C_r \gamma^{r-1}.$$

under weak assumptions: centered dissipativity at the target, and a linear-growth condition on the stochastic gradient. In particular, the objective may be nonconvex and the gradient need not be globally Lipschitz. This result gives the basic stationary scale of the nonlinear recursion in the infinite-variance regime and shows that the initial condition is forgotten exponentially fast.

Linear one-big-jump tail with exact constant. For the linearized recursion

$$Y_n = (I_d - \gamma H)Y_{n-1} + \gamma \eta_n, \quad \eta_n = -g(x^*, \varepsilon_n),$$

we first establish a general vector one-big-jump theorem under multivariate regular variation. We then specialize to linear functionals and, in a stable left eigen-direction, prove a projected tail formula under only a one-dimensional projected tail assumption.

Nonlinear transfer and exact projected tail probability. We prove that the nonlinear error $x_n - y_n$ is negligible at the rare-event scale. Combining this with the vector one-big-jump theorem yields exact tail asymptotics (3) for arbitrary linear functionals under multivariate regular variation. A left-eigendirection result under only projected tail regular variation is obtained as a special case.

Numerical calibration of the sharp constant. Our simulations compare empirical tail probabilities with the exact one-big-jump benchmark, rather than only checking slopes or upper bounds. Across a smooth nonconvex drift, a nonsmooth threshold autoregressive model, and expectile SGD with Student- t innovations, the empirical-to-theoretical tail ratio approaches one, supporting the sharpness of the predicted constant.

1.2 Related literature

SGD asymptotics and uncertainty quantification. Classical stochastic approximation traces back to Robbins and Monro [1951], with asymptotic efficiency of averaged recursions developed by Polyak and Juditsky [1992] and in the stochastic approximation monograph of Kushner and Yin [2003]. Non-asymptotic convergence theory for stochastic approximation and SGD is now extensive; see, for example, Moulines and Bach [2011]. More closely related to the inferential motivation of this paper, Chen et al. [2020] develop statistical inference for model parameters computed by SGD, Fang et al. [2018] propose online bootstrap confidence intervals, Yu et al. [2021] analyze constant-stepsizes SGD inference in nonconvex regimes, and Sheshukova et al. [2025] establish Gaussian approximation and multiplier bootstrap validity for averaged SGD, while Wei et al. [2025] develop Gaussian approximation and concentration results for constant-learning-rate SGD. These works are primarily finite-variance and Gaussian in nature.

Heavy-tailed SGD and stochastic optimization. Heavy-tailed stochastic gradient noise has been observed and modeled in modern learning problems [Simsekli et al., 2019, Gürbüzbalaban et al., 2021]. Theoretical work has studied convergence rates and limiting distributions of SGD under infinite-variance noise [Wang et al., 2021, Blanchet et al., 2024], as well as robust or clipped methods that achieve high-probability guarantees under weak moment assumptions [Gorbunov et al., 2020, Cutkosky and Mehta, 2021, Nguyen et al., 2023]. These results show that optimization remains possible under heavy-tailed noise and provide convergence or high-probability control.

Extreme value theory and regular variation. The mathematical mechanism behind our result is the one-big-jump principle for regularly varying random variables. Regular variation and Potter-type bounds [Bingham et al., 1987] are classical tools in extreme value theory [Bingham et al., 1987, Embrechts et al., 1997]. Tail asymptotics for heavy-tailed linear processes, infinite weighted series, and stochastic recurrence equations have been developed in, among others, Hult and Samorodnitsky [2008] and Mikosch and Wintenberger [2013]. Our linearized recursion is related to these weighted-sum settings, but the learning-rate-dependent weights and the nonlinear SGD transfer require additional analysis. In particular, the sharp constant in (3) arises from a Riemann-sum limit of the exponentially stable impulse response of the linearized dynamics.

Nonconvex and dissipative constant-stepsizes dynamics. Constant-stepsizes SGD can be viewed as a Markov chain, and this viewpoint has been used to study asymptotic expansions, bias, and stationary behavior [Dieuleveut et al., 2020, Yu et al., 2021]. Dissipativity assumptions are also standard in nonconvex sampling and learning dynamics, including analyses of stochastic gradient Langevin dynamics [Raginsky et al., 2017, Zhang et al., 2023]. Our assumptions follow this stability perspective but target a different phenomenon: rather than Gaussian fluctuations or stationary bias, we quantify the exact heavy-tailed rare-event probability of the last iterate.

1.3 Notations

For a vector $z = (z_1, \dots, z_d) \in \mathbb{R}^d$ and $q \geq 1$, we denote $|z|_q = (\sum_{i=1}^d |z_i|^q)^{1/q}$. We write $|z| = |z|_2$ for its Euclidean norm. For a matrix $B \in \mathbb{R}^{d \times d}$, $\|B\|_{\text{op}}$ denotes the operator norm. For a random vector $Z \in \mathbb{R}^d$ and $s \geq 1$, we write $\|Z\|_s := (\mathbb{E}|Z|^s)^{1/s}$ and say $Z \in \mathcal{L}^s$ if $\|Z\|_s < \infty$. Throughout this paper, all constants may change from line to line and may depend on fixed model parameters, but not on n or γ . The exponent $t \in (1, 2)$ denotes the regular-variation index of the

innovation $\eta = -g(x^*, \varepsilon)$; in the projected-tail version, the same notation is used for the tail index of $\eta^{(v)}$. The exponent $p \in (1, 2)$ denotes the largest moment of the stochastic gradient used in the nonlinear analysis; in the final heavy-tailed theorem we take $p \leq t$ when this moment is available.

2 Moment Convergence under Weak Local Assumptions

In this section, we establish a moment convergence theorem for the constant-stepsizes SGD iterates in (2). The assumptions below are deliberately local around the target except for a global linear-growth condition and a global centered dissipativity condition. In particular, the objective may be nonconvex away from the target and the stochastic gradient need not be globally Lipschitz.

Let

$$m(x) := \nabla M(x) = \mathbb{E}[g(x, \varepsilon)], \quad x \in \mathbb{R}^d \quad (4)$$

be the population gradient. We first introduce assumptions on the population gradient m and the stochastic gradient g ; see [Yu et al., 2021, Lou et al., 2022] for similar conditions.

Assumption 2.1 (Centered dissipativity). *There exists a constant $\alpha > 0$ such that*

$$\langle x - x^*, m(x) \rangle \geq \alpha |x - x^*|^2, \quad \forall x \in \mathbb{R}^d. \quad (5)$$

Assumption 2.1 is a one-point monotonicity condition centered at the target x^* . It is weaker than global strong convexity, as it does not require monotonicity between arbitrary pairs of points, but it is stronger than a purely tail dissipativity condition since (5) is imposed globally. This type of stability condition is common in the analysis of stochastic approximation and dissipative dynamical systems, and it is compatible with many nonconvex statistical objectives [Tan and Vershynin, 2023].

Assumption 2.2 (Regularity of the stochastic gradient). *There exist constants $L_p < \infty$, $L_g < \infty$, and $\kappa > 0$ such that, for some $1 < p < 2$,*

$$\|g(x, \varepsilon)\|_p \leq L_p(1 + |x - x^*|), \quad \forall x \in \mathbb{R}^d, \quad (6)$$

and

$$\|g(x, \varepsilon) - g(x^*, \varepsilon)\|_p \leq L_g |x - x^*|, \quad |x - x^*| \leq \kappa. \quad (7)$$

The global condition (6) is only a linear-growth condition in L^p , while (7) is imposed only near x^* . Thus we do not assume a globally Lipschitz gradient. By Jensen's inequality, (6) also implies linear growth of the population gradient m ; see Lemma C.1.

Assumption 2.3 (Local quadratic growth). *The gradient m is differentiable at x^* . Write*

$$H := \nabla m(x^*). \quad (8)$$

There exist constants $C_R < \infty$ and a radius $\kappa > 0$ (chosen no larger than the radius in Assumption 2.2, if necessary) such that

$$|m(x) - H(x - x^*)| \leq C_R |x - x^*|^2, \quad |x - x^*| \leq \kappa. \quad (9)$$

Remark 2.4. Condition (9) is implied, for example, by the local $C^{1,1}$ condition

$$\|\nabla m(x) - H\|_{\text{op}} \leq L_H |x - x^*|, \quad |x - x^*| \leq \kappa.$$

The following theorem gives the moment convergence of the SGD recursion. Its proof is given in Appendix A. Notably, this result uses only Assumption 2.1 and the global growth condition (6); the local Lipschitz and local quadratic growth conditions are not needed until the nonlinear transfer step.

Theorem 2.5 (L^r moment convergence for $1 < r \leq p$). *Assume Assumption 2.1 and the global growth condition (6) in Assumption 2.2. For every $r \in (1, p]$, there exist constants $C_r, c_r, \gamma_r > 0$ such that, for all $0 < \gamma < \gamma_r$ and all $n \geq 0$,*

$$\mathbb{E}|x_n - x^*|^r \leq (1 - c_r \gamma)^n |x_0 - x^*|^r + C_r \gamma^{r-1}. \quad (10)$$

Consequently,

$$\mathbb{P}(|x_n - x^*| > u) \leq C_p \frac{\gamma^{p-1}}{u^p} + C_p \frac{e^{-c_p n \gamma} |x_0 - x^*|^p}{u^p}, \quad u > 0. \quad (11)$$

Remark 2.6. The moment bound (10) shows that constant-stepsizes SGD forgets its initial condition exponentially fast on the time scale $1/\gamma$. The stationary contribution has order γ^{r-1} , which is the natural heavy-tailed analogue of the usual $\gamma^{r/2}$ finite-variance scale. In particular, when only moments below order two are available, the central-limit scaling is no longer the correct mechanism for the iterate tails.

3 Linear One-Big-Jump Exact Tails in the Infinite-variance Regime

The goal of this section is to identify the rare-event mechanism of constant-stepsizes SGD in the infinite-variance regime. We first state a general vector one-big-jump theorem for the linearized recursion under multivariate regular variation, which is then specialized to linear functionals. Finally, for a left eigen-direction of the linearized drift, the projection becomes a scalar autoregression; in that special case the multivariate regular variation assumption can be replaced by a simple one-dimensional right-tail regular variation condition together with a comparable left-tail condition.

Define the centered nonlinear iterate $X_n := x_n - x^*$. At the target, set

$$\eta_n := -g(x^*, \varepsilon_n). \quad (12)$$

Since $(\varepsilon_n)_{n \geq 1}$ are i.i.d., $(\eta_n)_{n \geq 1}$ are i.i.d. Moreover, because x^* is an unconstrained minimizer and $m(x^*) = 0$,

$$\mathbb{E}\eta_n = -\mathbb{E}[g(x^*, \varepsilon_n)] = -m(x^*) = 0. \quad (13)$$

We analyze the centered linearized recursion

$$Y_n = B_\gamma Y_{n-1} + \gamma \eta_n, \quad B_\gamma := I_d - \gamma H, \quad Y_0 = x_0 - x^*. \quad (14)$$

Iterating gives

$$Y_n = B_\gamma^n Y_0 + \gamma T_{\gamma,n}, \quad T_{\gamma,n} := \sum_{j=0}^{n-1} B_\gamma^j \eta_{n-j}. \quad (15)$$

Assumption 3.1 (Multivariate regularly varying innovation). *There exist $t \in (1, 2)$ and a non-null Radon measure μ_η on $\mathbb{R}^d \setminus \{0\}$ such that*

$$\frac{\mathbb{P}(r^{-1}\eta \in \cdot)}{\mathbb{P}(|\eta| > r)} \xrightarrow{v} \mu_\eta(\cdot), \quad r \rightarrow \infty, \quad (16)$$

where $\xrightarrow{v}$ denotes vague convergence on $\mathbb{R}^d \setminus \{0\}$. The measure μ_η is homogeneous of order $-t$: $\mu_\eta(aA) = a^{-t}\mu_\eta(A)$ for $a > 0$.

Assumption 3.2 (Shrinking-threshold diagonal regime). *As $\gamma \downarrow 0$, $n = n_\gamma \rightarrow \infty$ and $u_\gamma \downarrow 0$ satisfy*

$$n_\gamma \gamma \rightarrow \infty, \quad r_\gamma := \frac{u_\gamma}{\gamma} \rightarrow \infty, \quad \frac{\mathbb{P}(|\eta_0| > r_\gamma)}{\gamma} \rightarrow 0, \quad \frac{e^{-\alpha n_\gamma \gamma/2}}{u_\gamma} \rightarrow 0. \quad (17)$$

For a Borel set $A \subset \mathbb{R}^d \setminus \{0\}$, define

$$\nu_H(A) := \int_0^\infty \mu_\eta(\{z \in \mathbb{R}^d : e^{-sH} z \in A\}) ds = \int_0^\infty \mu_\eta \circ (e^{-sH})^{-1}(A) ds. \quad (18)$$

If A is bounded away from the origin, then $\nu_H(A) < \infty$ by Lemma B.1 and the homogeneity of μ_η .

Theorem 3.3 (Exact tail probability for the linearized vector recursion). *Assume Assumption 2.1, differentiability of m at x^* , Assumption 3.1 and 3.2. Let $A \subset \mathbb{R}^d \setminus \{0\}$ be a Borel set such that*

$$\text{dist}(A, 0) > 0, \quad \nu_H(\partial A) = 0, \quad \nu_H(A) > 0. \quad (19)$$

Then

$$\mathbb{P}(u_\gamma^{-1} Y_{n_\gamma} \in A) \sim \mathbb{P}(u_\gamma^{-1} \gamma T_{\gamma, n_\gamma} \in A) \sim \frac{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)}{\gamma} \nu_H(A). \quad (20)$$

Equivalently, in the one-big-jump ratio form,

$$\left| \frac{\mathbb{P}(u_\gamma^{-1} Y_{n_\gamma} \in A)}{\sum_{j=0}^{n_\gamma-1} \mathbb{P}(r_\gamma^{-1} B_\gamma^j \eta_0 \in A)} - 1 \right| \rightarrow 0. \quad (21)$$

Remark 3.4 (Typical choices of the rare-event set). *Theorem 3.3 is stated for a general ν_H -continuity set A bounded away from the origin. This set determines which rare event of the rescaled vector $u_\gamma^{-1} Y_{n_\gamma}$ is being measured. The most common choices are half-spaces, norm-exceedance sets, cones, and quadratic-exceedance sets. For instance,*

$$A_\ell := \{x \in \mathbb{R}^d : \ell^\top x > 1\}$$

corresponds exactly to the projected tail event

$$u_\gamma^{-1} Y_{n_\gamma} \in A_\ell \iff \ell^\top Y_{n_\gamma} > u_\gamma.$$

Similarly, $A_{\text{norm}} := \{x : |x| > 1\}$ gives the norm tail $|Y_{n_\gamma}| > u_\gamma$, while cone-type sets

$$A_C := \{x : |x| > 1, x/|x| \in C\}, \quad C \subseteq S^{d-1},$$

describe large deviations with prescribed directions. Thus the abstract constant $\nu_H(A)$ in Theorem 3.3 becomes an explicit tail constant once the shape of the rare event is chosen.

The next corollary uses the half-space choice $A = A_\ell$ to obtain the exact tail probability of an arbitrary linear functional.

Corollary 3.5 (Linear-functional tail under multivariate regular variation). *Assume the conditions of Theorem 3.3. Let $\ell \in \mathbb{R}^d \setminus \{0\}$ and define $A_\ell := \{x \in \mathbb{R}^d : \ell^\top x > 1\}$. Set*

$$C_\ell := \nu_H(A_\ell) = \int_0^\infty \mu_\eta \{z : \ell^\top e^{-sH} z > 1\} ds. \quad (22)$$

If $C_\ell > 0$, then

$$\mathbb{P}(\ell^\top Y_{n_\gamma} > u_\gamma) \sim \frac{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)}{\gamma} C_\ell. \quad (23)$$

For a general linear functional $\ell^\top Y_n$, the direction $(B_\gamma^j)^\top \ell$ changes with j , so the full multivariate tail measure is the natural object. In contrast, if v is a left eigenvector of H , the projected recursion is genuinely scalar. This allows us to replace Assumption 3.1 by a one-dimensional tail assumption on the projected innovation.

Specifically, let $v \in \mathbb{R}^d \setminus \{0\}$ be a fixed unit vector satisfying

$$H^\top v = \lambda_v v, \quad \lambda_v > 0. \quad (24)$$

Define the original-space linearized iterate $y_n := x^* + Y_n$. Set

$$y_n^{(v)} := \langle v, y_n - x^* \rangle = \langle v, Y_n \rangle, \quad \eta_n^{(v)} := \langle v, \eta_n \rangle = -\langle v, g(x^*, \varepsilon_n) \rangle. \quad (25)$$

Then $\mathbb{E}\eta_n^{(v)} = 0$, and by (24), $y_n^{(v)} = (1 - \gamma\lambda_v)y_{n-1}^{(v)} + \gamma\eta_n^{(v)}$. So the centered projected linear iterate is

$$S_n^{(v)} := y_n^{(v)} - \rho_{\gamma,v}^n y_0^{(v)} = \gamma \sum_{j=0}^{n-1} \rho_{\gamma,v}^j \eta_{n-j}^{(v)}, \quad y_0^{(v)} := \langle v, x_0 - x^* \rangle, \quad (26)$$

where $\rho_{\gamma,v} = 1 - \gamma\lambda_v$. Here we also define one-big-jump intensity

$$\Lambda_\gamma(u) := \sum_{j=0}^{n_\gamma-1} P\left(\gamma \rho_{\gamma,v}^j \eta_0^{(v)} > u\right), \quad u > 0. \quad (27)$$

Assumption 3.6 (Projected tail regular variation). *Let $v \in \mathbb{R}^d \setminus \{0\}$ be fixed and define*

$$\bar{F}_v^+(u) := \mathbb{P}(\eta_0^{(v)} > u), \quad \bar{F}_v^-(u) := \mathbb{P}(\eta_0^{(v)} < -u). \quad (28)$$

There exist $t \in (1, 2)$ and a slowly varying function h_v such that

$$\bar{F}_v^+(u) \sim h_v(u) u^{-t}, \quad u \rightarrow \infty. \quad (29)$$

Moreover, the left tail is dominated by the right-tail scale: there exist constants $C_- < \infty$ and $u_- < \infty$ such that

$$\bar{F}_v^-(u) \leq C_- \bar{F}_v^+(u), \quad u \geq u_-. \quad (30)$$

Theorem 3.7 (Projected linear exact tail). *Assume the left-eigenvector condition (24) and Assumption 3.6. Let $\gamma \downarrow 0$, $n = n_\gamma \rightarrow \infty$, and $u_\gamma \downarrow 0$ satisfy*

$$0 < \gamma\lambda_v < 1, \quad n_\gamma \gamma \rightarrow \infty, \quad r_\gamma := \frac{u_\gamma}{\gamma} \rightarrow \infty, \quad \frac{\bar{F}_v^+(r_\gamma)}{\gamma} \rightarrow 0, \quad \frac{\rho_{\gamma,v}^{n_\gamma}}{u_\gamma} \rightarrow 0. \quad (31)$$

Then

$$\mathbb{P}\left(y_{n_\gamma}^{(v)} > u_\gamma\right) \sim \Lambda_\gamma(u_\gamma) \sim \frac{1}{t\lambda_v} \frac{\bar{F}_v^+(u_\gamma/\gamma)}{\gamma} \sim \frac{1}{t\lambda_v} h_v(u_\gamma/\gamma) \gamma^{t-1} u_\gamma^{-t}. \quad (32)$$

Equivalently,

$$\left| \frac{\mathbb{P}(y_{n_\gamma}^{(v)} > u_\gamma)}{\Lambda_\gamma(u_\gamma)} - 1 \right| \rightarrow 0. \quad (33)$$

Remark 3.8. The equation 33 admits a constructive result: the linear tail probability is asymptotically equivalent to a sum of marginal single-jump probabilities. The closed-form sharp constant C_l is recovered by passing the Riemann-sum limit after replacing $\bar{F}_v^+$ with its regular-variation asymptotic: when an exact distribution function is available, one may evaluate the one-jump sum directly.

Remark 3.9 (Relation between the vector and projected assumptions). Under the stronger Assumption 3.1, if $A_v := \{z : v^\top z > 1\}$ satisfies $\mu_\eta(A_v) > 0$, then the projected right-tail regular variation (29) follows with $h_v(u)u^{-t} \sim \mu_\eta(A_v)\mathbb{P}(|\eta_0| > u)$. However, the projected theorem above only needs the one-dimensional right tail of $\eta_0^{(v)}$ and the comparable left-tail bound (30).

Remark 3.10 (Power thresholds). For $u_\gamma = \gamma^a$ with $0 < a < 1$, the rare-event part of (31) becomes

$$h_v(\gamma^{a-1})\gamma^{t-1-at} \rightarrow 0. \quad (34)$$

The initialization part is implied, for example, by $n_\gamma\gamma/\log(1/\gamma) \rightarrow \infty$. Under these conditions,

$$\mathbb{P}(y_{n_\gamma}^{(v)} > \gamma^a) \sim \frac{1}{t\lambda_v} h_v(\gamma^{a-1})\gamma^{t-1-at}. \quad (35)$$

If $h_v(u) \rightarrow h_{v,0} \in (0, \infty)$, then (34) is equivalent to $a < (t-1)/t$.

4 Nonlinear Transfer and Exact Tail Asymptotics

We now transfer the projected tail from Section 3 to the raw SGD iterate. For $\ell \in \mathbb{R}^d \setminus \{0\}$, write

$$\Delta_n := x_n - y_n, \quad \Delta_n^{(\ell)} := \langle \ell, \Delta_n \rangle. \quad (36)$$

The nonlinear and linear projected errors are

$$x_n^{(\ell)} := \langle \ell, x_n - x^* \rangle, \quad y_n^{(\ell)} := \langle \ell, y_n - x^* \rangle. \quad (37)$$

The following theorem characterizes the nonlinear linearization error. Its proof is in Appendix C.

Theorem 4.1 (Linear approximation in L^q). Assume Assumptions 2.1, 2.2, and 2.3. Let q satisfy $\frac{p+1}{2} < q < p$. Then there exist constants $C_q, c_q, \gamma_q > 0$ such that, for all $0 < \gamma < \gamma_q$ and all $n \geq 1$,

$$\|\Delta_n\|_q^q \leq C_q \gamma^{p-1} + C_q e^{-c_q n \gamma} (1 + |x_0 - x^*|^p). \quad (38)$$

We now state the final transfer results for general linear functionals with power thresholds.

Theorem 4.2 (Exact nonlinear tail asymptotics for general linear functionals). Assume Assumptions 2.1, 2.2, 2.3, and 3.1. Let $\ell \in \mathbb{R}^d \setminus \{0\}$ and assume the linear tail constant C_ℓ defined in (22) is positive. Suppose the moment exponent in Assumption 2.2 satisfies $p \in (1, t]$. Let $u_\gamma = \gamma^a$ for some constant a such that there exists some $q \in ((p+1)/2, p)$ that satisfies $0 < a < \frac{t-1}{t}, p-t+a(t-q) > 0$. Then

$$\mathbb{P}(x_{n_\gamma}^{(\ell)} > \gamma^a) \sim \mathbb{P}(y_{n_\gamma}^{(\ell)} > \gamma^a) \sim C_\ell \frac{\mathbb{P}(|\eta_0| > \gamma^{a-1})}{\gamma}, \text{ as } \gamma \downarrow 0 \text{ and } \frac{n_\gamma \gamma}{\log(1/\gamma)} \rightarrow \infty.$$

Corollary 4.3 (Left-eigendirection under a projected tail assumption). Assume Assumptions 2.1, 2.2, and 2.3. Let v be a unit left eigenvector of $\nabla m(x^*)$ and assume the projected tail condition in Assumption 3.6. Suppose the moment exponent in Assumption 2.2 satisfies $p \in (1, t]$. Let $u_\gamma = \gamma^a$ for some constant a such that there exists some $q \in ((p+1)/2, p)$ that satisfies $0 < a < \frac{t-1}{t}, p-t+a(t-q) > 0$. Then, as $\gamma \downarrow 0$ and $\frac{n_\gamma \gamma}{\log(1/\gamma)} \rightarrow \infty$, we have,

$$\mathbb{P}(x_{n_\gamma}^{(v)} > \gamma^a) \sim \mathbb{P}(y_{n_\gamma}^{(v)} > \gamma^a) \sim \Lambda_\gamma(u_\gamma) \sim \frac{1}{t\lambda_v} h_v(\gamma^{a-1})\gamma^{t-1-at}.$$

Equivalently,

$$\left| \frac{\mathbb{P}(x_{n_\gamma}^{(v)} > u_\gamma)}{\Lambda_\gamma(u_\gamma)} - 1 \right| \rightarrow 0. \quad (39)$$

Remark 4.4. Corollary 4.3 give the asymptotic equivalence between the projected nonlinear tail and its linearized counterpart at the rare-event scale. Combined with Remark 3.8, this extends to a three-way asymptotic equivalence with the linear one-jump sum, which depends only on the linearization H , the step size γ , and the marginal law of innovation η_0 , with no joint distribution of any iterate involved.

Remark 4.5 (Role of the eigendirection assumption). The nonlinear transfer step does not require v to be a left eigenvector. The eigendirection condition is only used in Corollary 4.3 to reduce the linearized projected recursion to a scalar autoregression, so that one only needs the one-dimensional projected tail assumption Assumption 3.6. Under the stronger multivariate regular variation assumption, the general constant in Theorem 4.2 reduces to the special form in Corollary 4.3, i.e.

$$C_v = \int_0^\infty \mu_\eta\{z : e^{-\lambda_v s} v^\top z > 1\} ds = \frac{\mu_\eta\{z : v^\top z > 1\}}{t\lambda_v}.$$

Remark 4.6. The condition $p - t + a(t - q) > 0$ is the transfer condition. It ensures that the nonlinear linearization error in Theorem 4.1 is negligible relative to the one-big-jump tail in Theorem 3.7. When $p = t$, the condition holds for every $a > 0$ and every $q < t$. This endpoint case is valid only when the p th moment required in Assumption 2.2 is finite. For standard regularly varying examples such as Student- t_ν innovations with tail index ν , one should take any $p < \nu$ rather than $p = \nu$.

5 Numerical Studies

5.1 Simulation setup

In this section, we conduct a simulation study on three representative stochastic recursions to examine the sharpness of the exact tail approximation in Theorem 4.2. Our goal is not merely to verify that the proposed tail expression gives an upper bound, but to check whether the empirical tail probability is well calibrated by the exact one-big-jump constant predicted by the linearized recursion.

Throughout the experiments, we consider constant-stepsize SGD recursions of the form

$$X_k = X_{k-1} - \gamma m(X_{k-1}) + \gamma \xi_k, \quad k \geq 1, \quad (40)$$

or, in the expectile experiment, the corresponding stochastic-gradient recursion defined by an asymmetric squared loss. The additive noise variables $(\xi_k)_{k \geq 1}$ are i.i.d. Student- t_ν random variables.

For accuracy, the one-big-jump benchmark in Corollary 4.3 is precisely Λ_γ defined in Equation 27. Since the Student- t_ν survival function is available numerically, we compute (27) using the exact Student- t_ν tail rather than replacing it immediately by its asymptotic power-law approximation.

Let $\hat{P}_X(y)$ denote the empirical tail probability of the SGD recursion. We consider the diagnostic ratios $R_\gamma(y) := \frac{\hat{P}_X(y)}{\Lambda_\gamma(y)}$. Note that the ratio reported here is much more accurate than the absolute difference, since $\Lambda_\gamma(y) \rightarrow 0$ as $y \rightarrow \infty$. The following three tasks are studied.

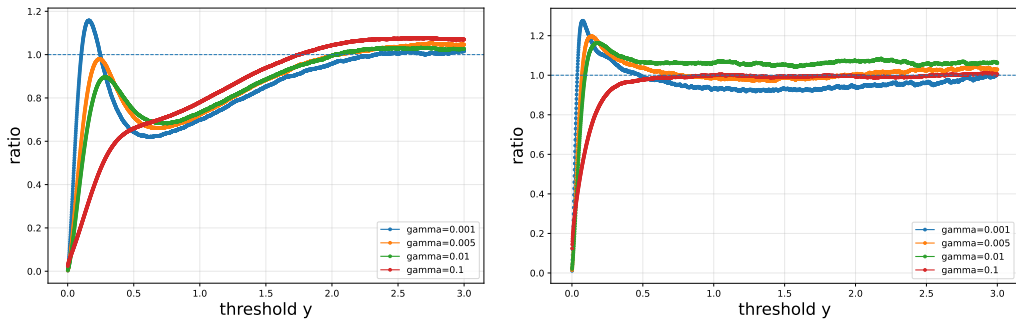


Figure 1: The empirical-to-theoretical tail ratio $R_\gamma(y)$ of the SGD recursion with Student $t_{1.5}$ innovations. Left: Task 1, the smooth nonconvex nonlinear drift. Right: Task 2, the threshold autoregressive drift.

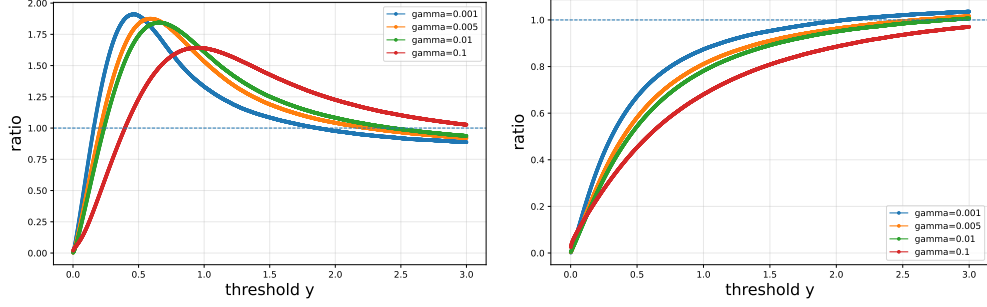


Figure 2: The empirical-to-theoretical tail ratio $R_\gamma(y)$ of the SGD recursion for Task 3, the expectile estimation with Student $t_{1.2}$ innovations. Left: $\tau = 0.4$. Right: $\tau = 0.6$.

Task 1: Smooth nonconvex nonlinear drift. The first task uses the nonconvex nonlinear drift

$$m(x) = x + 4 \frac{(\sin x)^3}{1 + x^2}. \quad (41)$$

The heavy-tailed distribution $t_{1.5}$ is chosen as the noise. This example is designed to represent a smooth nonconvex nonlinear recursion with heavy-tailed innovations.

Task 2: Threshold autoregressive drift. The second task considers a threshold-type piecewise-linear drift

$$m(x) = 3x \mathbf{1}_{\{x < 0\}} + 5x \mathbf{1}_{\{x \geq 0\}}. \quad (42)$$

The heavy-tailed distribution $t_{1.5}$ is chosen as the noise. Note that the associated objective is strongly convex but has a kink at the target. Thus this task serves as a nonsmooth threshold stress test and is directly related to the popular time series model threshold-AR(1).

Task 3: Expectile SGD. The third task is a statistical estimation problem based on expectiles. For $\tau \in (0, 1)$, the τ -expectile of a random variable Z is the minimizer of the asymmetric squared loss

$$\theta_\tau^* = \arg \min_{\theta \in \mathbb{R}} \mathbb{E} [|\tau - \mathbf{1}_{\{Z - \theta \leq 0\}}| (Z - \theta)^2]. \quad (43)$$

The stochastic gradient is

$$g(\theta, Z) = 2 [\tau \mathbf{1}_{\{Z > \theta\}} + (1 - \tau) \mathbf{1}_{\{Z \leq \theta\}}] (\theta - Z). \quad (44)$$

We consider the centered recursion

$$X_k = X_{k-1} - \gamma g(\theta_\tau^* + X_{k-1}, Z_k), \quad (45)$$

where Z_k follows a Student- $t_{1.2}$ distribution. We report results for two levels, $\tau = 0.4$ and $\tau = 0.6$.

5.2 Numerical results

We run 100 independent trials with $n = 10^6$ and $\gamma \in \{0.1, 0.01, 0.005, 0.001\}$. The range of the tail threshold y is chosen as $[0.001, 3.0]$. Figure 1 and 2 show how the ratios $R_\gamma(y)$ changes as the tail threshold y increases. For the smooth nonlinear drift and the expectile recursion, the ratios support the exact constant predicted by our theory. Although the threshold autoregressive drift is nonsmooth at the target, interestingly, the same one-big-jump benchmark remains well calibrated empirically. Moreover, as γ approaches 0, empirically the minimal y that is required to enter the asymptotic regime also decreases, as explicitly shown in Figure 2. This echoes with our infinitesimal regime where $\gamma \rightarrow 0$ and $n_\gamma \gamma \rightarrow \infty$ and is consistent with Theorem 4.2. In conclusion, the experimental results suggest that our exact tail probability approximation for SGD iterates with heavy-tailed gradient noises is valid and tight, precisely describing the tail behavior of constant learning rate SGD.

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A Proof of the L^p Moment Bound

We use the following elementary inequality for $1 < r \leq 2$.

Lemma A.1 (Lemma A.1 in Li et al. [2024]). *Let $u \in \mathbb{R}^d$ and $w \in \mathbb{R}^d$ be two random vectors such that $\mathbb{E}|u|^q < \infty$ and $\mathbb{E}|w|^q < \infty$ for some constant $q \in (1, 2)$. Then, we have*

$$\mathbb{E}|u + w|^q \leq \mathbb{E}|u|^q + q\mathbb{E}\left\{|u|^{q-2}u^\top \mathbb{E}(w|u)\right\} + 2^{2-q}\mathbb{E}|w|^q \quad (46)$$

Proof of Theorem 2.5. Fix $r \in (1, p]$. Since (6) holds in L^p , monotonicity of L^s norms gives

$$\|g(x, \varepsilon)\|_r \leq L_p(1 + |x - x^*|), \quad x \in \mathbb{R}^d. \quad (47)$$

Applying Lemma A.1 conditionally on $\mathcal{F}_{n-1} := \sigma(\varepsilon_1, \dots, \varepsilon_{n-1})$ with $u = x_{n-1} - x^*$ and $w = -\gamma g(x_{n-1}, \varepsilon_n)$ yields

$$\begin{aligned} \mathbb{E}[|x_n - x^*|^r \mid \mathcal{F}_{n-1}] &\leq |x_{n-1} - x^*|^r - r\gamma |x_{n-1} - x^*|^{r-2} \langle x_{n-1} - x^*, m(x_{n-1}) \rangle \\ &\quad + C_r \gamma^r \mathbb{E}[|g(x_{n-1}, \varepsilon_n)|^r \mid \mathcal{F}_{n-1}]. \end{aligned} \quad (48)$$

By centered dissipativity (5),

$$|x_{n-1} - x^*|^{r-2} \langle x_{n-1} - x^*, m(x_{n-1}) \rangle \geq \alpha |x_{n-1} - x^*|^r. \quad (49)$$

By (47),

$$\mathbb{E}[|g(x_{n-1}, \varepsilon_n)|^r \mid \mathcal{F}_{n-1}] \leq C_r(1 + |x_{n-1} - x^*|^r). \quad (50)$$

Combining (48)–(50) gives

$$\mathbb{E}[|x_n - x^*|^r \mid \mathcal{F}_{n-1}] \leq (1 - r\alpha\gamma + C_r\gamma^r) |x_{n-1} - x^*|^r + C_r\gamma^r. \quad (51)$$

Since $r > 1$, for sufficiently small γ ,

$$1 - r\alpha\gamma + C_r\gamma^r \leq 1 - c_r\gamma$$

for some $c_r > 0$. Taking expectations and iterating yields

$$\begin{aligned} \mathbb{E}|x_n - x^*|^r &\leq (1 - c_r\gamma)^n |x_0 - x^*|^r + C_r\gamma^r \sum_{j=0}^{n-1} (1 - c_r\gamma)^j \\ &\leq (1 - c_r\gamma)^n |x_0 - x^*|^r + C_r\gamma^{r-1}. \end{aligned} \quad (52)$$

This proves (10). The tail bound (11) follows from (10) with $r = p$ and Markov's inequality. $\square$

B Proofs for the Linearized Recursion

B.1 Stability of the vector linearization

Lemma B.1 (Stability of the vector linearization). *Assume Assumption 2.1 and suppose that m is differentiable at x^* with $H = \nabla m(x^*)$. Then*

$$\frac{1}{2}(H + H^\top) \succeq \alpha I_d. \quad (53)$$

Consequently, H is positive stable. Moreover, there exists $\gamma_0 > 0$ such that, for all $0 < \gamma < \gamma_0$, $B_\gamma := I_d - \gamma H$ satisfies

$$\|B_\gamma^j\|_{\text{op}} \leq e^{-\alpha\gamma j/2}, \quad j \geq 0. \quad (54)$$

Proof of Lemma B.1. Fix $u \in \mathbb{R}^d$ and $r > 0$. Apply Assumption 2.1 with $x = x^* + ru$. Since $m(x^*) = 0$,

$$\left\langle u, \frac{m(x^* + ru) - m(x^*)}{r} \right\rangle \geq \alpha \|u\|^2.$$

Letting $r \downarrow 0$ gives $\langle u, Hu \rangle \geq \alpha \|u\|^2$. Since $\langle u, Hu \rangle = \langle u, (H + H^\top)u/2 \rangle$, this proves (53).

If $H z = \lambda z$ for a possibly complex eigenvector $z \neq 0$, then

$$\operatorname{Re}(\lambda) \|z\|^2 = \operatorname{Re}(z^* H z) = z^* \frac{H + H^\top}{2} z \geq \alpha \|z\|^2,$$

so H is positive stable. Moreover,

$$\|B_\gamma u\|^2 = \|u - \gamma H u\|^2 \leq (1 - 2\alpha\gamma + \gamma^2 \|H\|_{\text{op}}^2) \|u\|^2.$$

For sufficiently small γ , the right-hand side is bounded by $(1 - \alpha\gamma) \|u\|^2$. Hence

$$\|B_\gamma u\| \leq (1 - \alpha\gamma)^{1/2} \|u\| \leq e^{-\alpha\gamma/2} \|u\|.$$

Iterating proves (54). $\square$

B.2 Proof of Theorem 3.3

Throughout this subsection, write

$$Q_\gamma := \mathbb{P}(|\eta_0| > r_\gamma), \quad X_{\gamma,j} := B_\gamma^j \eta_{n_\gamma-j}, \quad 0 \leq j < n_\gamma.$$

Then $T_{\gamma, n_\gamma} = \sum_{j=0}^{n_\gamma-1} X_{\gamma,j}$. By Lemma B.1, there exists $c_B > 0$ such that, for all sufficiently small γ ,

$$\|B_\gamma^j\|_{\text{op}} \leq e^{-c_B \gamma j}, \quad j \geq 0. \quad (55)$$

Lemma B.2 (One-jump Riemann limit). *Let $A \subset \mathbb{R}^d \setminus \{0\}$ be a Borel set satisfying (19). Then*

$$\gamma \sum_{j=0}^{n_\gamma-1} \frac{\mathbb{P}(r_\gamma^{-1} B_\gamma^j \eta_0 \in A)}{Q_\gamma} \longrightarrow \nu_H(A). \quad (56)$$

The same conclusion holds with A replaced by any ν_H -continuity set bounded away from the origin.

Proof. Fix $K < \infty$. Since $n_\gamma \gamma \rightarrow \infty$, the grid $\{j : \gamma j \leq K\}$ is eventually contained in $\{0, \dots, n_\gamma - 1\}$. On $0 \leq \gamma j \leq K$,

$$\sup_{0 \leq j \leq K/\gamma} \|B_\gamma^j - e^{-\gamma j H}\|_{\text{op}} \rightarrow 0. \quad (57)$$

By product vague convergence and the continuous mapping $(s, z) \mapsto e^{-sH} z$,

$$\gamma \sum_{0 \leq j \leq K/\gamma} \frac{\mathbb{P}(r_\gamma^{-1} e^{-\gamma j H} \eta_0 \in A)}{Q_\gamma} \longrightarrow \int_0^K \mu_\eta \{z : e^{-sH} z \in A\} ds, \quad (58)$$

where the boundary condition follows from $\nu_H(\partial A) = 0$. The replacement of $e^{-\gamma j H}$ by B_γ^j follows from (57) and the usual inner/outer enlargement argument; because A is bounded away from the origin, the contribution of $|z| > R$ is controlled uniformly by Potter's bound [Resnick, 1987] and then sent to zero as $R \rightarrow \infty$. Hence

$$\gamma \sum_{0 \leq j \leq K/\gamma} \frac{\mathbb{P}(r_\gamma^{-1} B_\gamma^j \eta_0 \in A)}{Q_\gamma} \longrightarrow \int_0^K \mu_\eta \{z : e^{-sH} z \in A\} ds. \quad (59)$$

It remains to let $K \rightarrow \infty$. If $r_\gamma^{-1} B_\gamma^j \eta_0 \in A$ and $a := \operatorname{dist}(A, 0) > 0$, then $|\eta_0| \geq a r_\gamma / \|B_\gamma^j\|_{\text{op}}$. Potter's bound [Resnick, 1987] and (55) imply that, for every sufficiently small $\zeta > 0$,

$$\frac{\mathbb{P}(r_\gamma^{-1} B_\gamma^j \eta_0 \in A)}{Q_\gamma} \leq C_A \|B_\gamma^j\|_{\text{op}}^{t-\zeta} \leq C_A e^{-(t-\zeta)c_B \gamma j}.$$

Therefore the discrete tail over $\gamma j > K$ is $O(e^{-cK})$, and the same bound controls the tail of the integral defining $\nu_H(A)$. Combining this with (59) proves (56). $\square$

Lemma B.3 (Negligibility of many small jumps). *For every $\delta > 0$,*

$$\lim_{\theta \downarrow 0} \limsup_{\gamma \downarrow 0} \frac{\gamma}{Q_\gamma} \mathbb{P} \left(\left\| \sum_{j=0}^{n_\gamma-1} X_{\gamma,j} \mathbf{1}_{\{\|X_{\gamma,j}\| \leq \theta r_\gamma\}} \right\| > \delta r_\gamma \right) = 0. \quad (60)$$

Proof. Let $\tilde{X}_{\gamma,j}^{(\theta)} := X_{\gamma,j} \mathbf{1}_{\{\|X_{\gamma,j}\| \leq \theta r_\gamma\}}$ and $\bar{X}_{\gamma,j}^{(\theta)} := \tilde{X}_{\gamma,j}^{(\theta)} - \mathbb{E} \tilde{X}_{\gamma,j}^{(\theta)}$. Choose $\beta \in (t, 2)$ and $0 < \zeta < \beta - t$. A coordinate-wise von Bahr–Esseen inequality, equivalence of norms in finite dimension, and independence give

$$\mathbb{E} \left\| \sum_{j=0}^{n_\gamma-1} \bar{X}_{\gamma,j}^{(\theta)} \right\|^\beta \leq C_\beta \sum_{j=0}^{n_\gamma-1} \mathbb{E} \|\tilde{X}_{\gamma,j}^{(\theta)}\|^\beta.$$

By Karamata's theorem [Resnick, 1987] for the norm tail, Potter's bound [Resnick, 1987], and (55),

$$\sum_{j=0}^{n_\gamma-1} \mathbb{E} \|\tilde{X}_{\gamma,j}^{(\theta)}\|^\beta \leq C r_\gamma^\beta Q_\gamma \theta^{\beta-t-\zeta} \sum_{j=0}^{n_\gamma-1} \|B_\gamma^j\|_{\text{op}}^{t-\zeta} \leq C r_\gamma^\beta Q_\gamma \theta^{\beta-t-\zeta} \gamma^{-1}.$$

Markov's inequality yields

$$\mathbb{P} \left(\left\| \sum_{j=0}^{n_\gamma-1} \bar{X}_{\gamma,j}^{(\theta)} \right\| > \frac{\delta r_\gamma}{2} \right) \leq C \delta^{-\beta} \theta^{\beta-t-\zeta} \frac{Q_\gamma}{\gamma}. \quad (61)$$

Since $\mathbb{E} X_{\gamma,j} = B_\gamma^j \mathbb{E} \eta_0 = 0$,

$$\mathbb{E} \tilde{X}_{\gamma,j}^{(\theta)} = -\mathbb{E} [X_{\gamma,j} \mathbf{1}_{\{\|X_{\gamma,j}\| > \theta r_\gamma\}}].$$

Again by Karamata's theorem and Potter's bound [Resnick, 1987],

$$\left\| \sum_{j=0}^{n_\gamma-1} \mathbb{E} \tilde{X}_{\gamma,j}^{(\theta)} \right\| \leq C r_\gamma \theta^{1-t-\zeta} \frac{Q_\gamma}{\gamma} = o(r_\gamma)$$

for each fixed $\theta > 0$, because $Q_\gamma/\gamma \rightarrow 0$. Combining this with (61) and then letting $\theta \downarrow 0$ proves the lemma. $\square$

Proof of Theorem 3.3. Let A satisfy (19), and set $a := \text{dist}(A, 0) > 0$. For $\varepsilon > 0$, define $A^\varepsilon := \{x : \text{dist}(x, A) < \varepsilon\}$ and $A_{-\varepsilon} := \{x \in A : \text{dist}(x, A^c) > \varepsilon\}$. Choose $\varepsilon < a/4$ so that A^ε and $A_{-\varepsilon}$ are ν_H -continuity sets.

For $\theta \in (0, a/4)$, let $N_{\gamma,\theta} := \sum_{j=0}^{n_\gamma-1} \mathbf{1}_{\{\|X_{\gamma,j}\| > \theta r_\gamma\}}$. Lemma B.2 applied to the norm-exceedance set gives $\mathbb{E} N_{\gamma,\theta} = O(Q_\gamma/\gamma)$; hence, by independence and $Q_\gamma/\gamma \rightarrow 0$,

$$\mathbb{P}(N_{\gamma,\theta} \geq 2) = o(Q_\gamma/\gamma). \quad (62)$$

For the upper bound, if $r_\gamma^{-1} \sum_j X_{\gamma,j} \in A$, if at most one summand is larger than θr_γ , and if the sum of the small summands has norm at most εr_γ , then the unique large summand, if present, must lie in $r_\gamma A^\varepsilon$. Thus

$$\begin{aligned} \mathbb{P} \left(r_\gamma^{-1} \sum_j X_{\gamma,j} \in A \right) &\leq \mathbb{P} \left(\bigcup_j \{r_\gamma^{-1} X_{\gamma,j} \in A^\varepsilon\} \right) + \mathbb{P}(N_{\gamma,\theta} \geq 2) \\ &\quad + \mathbb{P} \left(\left\| \sum_j X_{\gamma,j} \mathbf{1}_{\{\|X_{\gamma,j}\| \leq \theta r_\gamma\}} \right\| > \varepsilon r_\gamma \right). \end{aligned}$$

Using the union bound, Lemma B.2, (62), and Lemma B.3, then letting $\theta \downarrow 0$ and $\varepsilon \downarrow 0$, gives

$$\limsup_{\gamma \downarrow 0} \frac{\gamma}{Q_\gamma} \mathbb{P} \left(r_\gamma^{-1} \sum_j X_{\gamma,j} \in A \right) \leq \nu_H(A).$$

The lower bound is analogous. On the event that one summand lies in $r_\gamma A_{-\varepsilon}$, no second summand is larger than θr_γ , and the small-summand contribution has norm at most εr_γ , the full sum lies in $r_\gamma A$. Bonferroni's inequality and the same estimates yield

$$\liminf_{\gamma \downarrow 0} \frac{\gamma}{Q_\gamma} \mathbb{P} \left(r_\gamma^{-1} \sum_j X_{\gamma,j} \in A \right) \geq \nu_H(A).$$

Therefore

$$\mathbb{P}(r_\gamma^{-1}T_{\gamma,n_\gamma} \in A) = \frac{Q_\gamma}{\gamma}\{\nu_H(A) + o(1)\}. \quad (63)$$

Since $r_\gamma = u_\gamma/\gamma$ and $\nu_H(A) > 0$, this is exactly the second asymptotic of (20). Lemma B.2 also gives the corresponding one-big-jump intensity approximation

$$\sum_{j=0}^{n_\gamma-1} \mathbb{P}(r_\gamma^{-1}B_\gamma^j\eta_0 \in A) \sim \frac{Q_\gamma}{\gamma}\nu_H(A),$$

which is used in the proof but is not needed in the statement.

Finally, by (15), $Y_{n_\gamma} = B_\gamma^{n_\gamma}Y_0 + \gamma T_{\gamma,n_\gamma}$. Assumption 3.2 and Lemma B.1 imply $u_\gamma^{-1}B_\gamma^{n_\gamma}Y_0 \rightarrow 0$. A final inner/outer enlargement argument transfers the second asymptotic of (20) to Y_{n_γ} and proves the first asymptotic of (20). $\square$

B.3 Proof of Corollary 3.5

Proof. For $c > 0$, let $A_{\ell,c} := \{x : \ell^\top x > c\}$. This set is bounded away from the origin. Its boundary is the affine hyperplane $\partial A_{\ell,c} = \{x : \ell^\top x = c\}$. For each $s \geq 0$,

$$(e^{-sH})^{-1}\partial A_{\ell,c} = \{z : \ell^\top e^{-sH}z = c\},$$

which is an affine hyperplane not passing through the origin.

We claim that a homogeneous Radon measure on $\mathbb{R}^d \setminus \{0\}$ assigns zero mass to such a hyperplane. It is enough to check this on annuli $\{r \leq \|z\| \leq R\}$. If an affine hyperplane L had positive μ_η -mass on such an annulus, then for infinitely many distinct λ_m sufficiently close to one, the sets $\lambda_m(L \cap \{r \leq \|z\| \leq R\})$ would be disjoint, have uniformly positive μ_η -mass by homogeneity, and be contained in a slightly larger compact annulus away from zero. This contradicts Radon finiteness. Hence $\mu_\eta\{z : \ell^\top e^{-sH}z = c\} = 0$ for every $s \geq 0$, and therefore $\nu_H(\partial A_{\ell,c}) = 0$.

Taking $c = 1$ and applying Theorem 3.3 to A_ℓ gives

$$\mathbb{P}(\ell^\top \gamma T_{\gamma,n_\gamma} > u_\gamma) \sim \frac{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)}{\gamma} C_\ell.$$

The deterministic shift $\ell^\top B_\gamma^{n_\gamma}Y_0$ is $o(u_\gamma)$ by Assumption 3.2. Applying the same result to $(1 \pm \varepsilon)u_\gamma$ and then letting $\varepsilon \downarrow 0$ yields (23). $\square$

B.4 Proof of Theorem 3.7

Proof. For notational simplicity, write $u = u_\gamma$, $n = n_\gamma$, $\rho = \rho_{\gamma,v} = 1 - \gamma\lambda_v$, $\lambda = \lambda_v$, and $\eta_j = \eta_j^{(v)}$. Put

$$Z_j := \gamma\rho^j\eta_{n-j}, \quad 0 \leq j < n,$$

so that $S_n^{(v)} = \sum_{j=0}^{n-1} Z_j$. Define the one-big-jump intensity

$$\Lambda_\gamma(u) := \sum_{j=0}^{n-1} \mathbb{P}(Z_j > u) = \sum_{j=0}^{n-1} \bar{F}_v^+(u/(\gamma\rho^j)).$$

We first establish the asymptotic

$$\Lambda_\gamma(u) \sim \frac{1}{t\lambda} \frac{\bar{F}_v^+(u/\gamma)}{\gamma}. \quad (64)$$

Since the multipliers ρ^{-j} range over an unbounded set as j varies in $\{0, \dots, n-1\}$, neither Potter's bound [Resnick, 1987] alone nor the uniform convergence theorem alone yields (64) with the sharp constant; we therefore split the sum at a cutoff $K > 0$ (assumed below to satisfy $K < n\gamma$, which holds eventually since $n\gamma \rightarrow \infty$). For $\gamma j \leq K$, the values ρ^{-j} lie in $[1, \rho^{-\lfloor K/\gamma \rfloor}]$, which converges to $[1, e^{\lambda K}]$ as $\gamma \downarrow 0$ and hence remains in a fixed compact subset of $[1, \infty)$ for small γ . Combining

$u/\gamma \rightarrow \infty$ (from (31)) with the uniform convergence theorem for slowly varying functions [Resnick, 1987] applied to h_v ,

$$\frac{\bar{F}_v^+(u/(\gamma\rho^j))}{\bar{F}_v^+(u/\gamma)} = \frac{h_v(u/(\gamma\rho^j))}{h_v(u/\gamma)} \rho^{tj} = (1 + o(1)) \rho^{tj}, \quad \gamma \downarrow 0,$$

uniformly in j with $\gamma j \leq K$. Using $1 - \rho^t \sim t\gamma\lambda$ and $\rho^{t\lfloor K/\gamma \rfloor} \rightarrow e^{-t\lambda K}$,

$$\lim_{\gamma \downarrow 0} \frac{\gamma}{\bar{F}_v^+(u/\gamma)} \sum_{j: \gamma j \leq K} \bar{F}_v^+(u/(\gamma\rho^j)) = \frac{1 - e^{-t\lambda K}}{t\lambda}. \quad (65)$$

For $\gamma j > K$ we use Potter's bound [Resnick, 1987]: fix $\varepsilon \in (0, t-1)$, then there exist $A_\varepsilon < \infty$ and $X_\varepsilon < \infty$ such that for all $x \geq X_\varepsilon$ and $\mu \geq 1$,

$$\bar{F}_v^+(\mu x)/\bar{F}_v^+(x) \leq A_\varepsilon \mu^{-t+\varepsilon}.$$

For γ small enough that $u/\gamma \geq X_\varepsilon$, taking $\mu = \rho^{-j}$ gives $\bar{F}_v^+(u/(\gamma\rho^j)) \leq A_\varepsilon \rho^{(t-\varepsilon)j} \bar{F}_v^+(u/\gamma)$ for every $j \geq 0$, and consequently

$$\limsup_{\gamma \downarrow 0} \frac{\gamma}{\bar{F}_v^+(u/\gamma)} \sum_{j: \gamma j > K} \bar{F}_v^+(u/(\gamma\rho^j)) \leq \frac{A_\varepsilon e^{-(t-\varepsilon)\lambda K}}{(t-\varepsilon)\lambda}. \quad (66)$$

Combining (65) and (66), for every fixed $K > 0$,

$$\frac{1 - e^{-t\lambda K}}{t\lambda} \leq \liminf_{\gamma \downarrow 0} \frac{\gamma \Lambda_\gamma(u)}{\bar{F}_v^+(u/\gamma)} \leq \limsup_{\gamma \downarrow 0} \frac{\gamma \Lambda_\gamma(u)}{\bar{F}_v^+(u/\gamma)} \leq \frac{1 - e^{-t\lambda K}}{t\lambda} + \frac{A_\varepsilon e^{-(t-\varepsilon)\lambda K}}{(t-\varepsilon)\lambda}.$$

Letting $K \rightarrow \infty$ with ε fixed proves (64). In particular, $\Lambda_\gamma(u) \rightarrow 0$ by (31). The same argument applied to cu in place of u , together with $\bar{F}_v^+(cu/\gamma)/\bar{F}_v^+(u/\gamma) \rightarrow c^{-t}$ (which is direct since cu/γ and u/γ are comparable up to a fixed ratio), yields the shift stability

$$\Lambda_\gamma(cu) \sim c^{-t} \Lambda_\gamma(u), \quad c > 0 \text{ fixed}. \quad (67)$$

We next show that $\mathbb{P}(S_n^{(v)} > u) \sim \Lambda_\gamma(u)$. Fix $\delta \in (0, 1/2)$ and $\theta \in (0, \delta/16)$, and define the truncated sum

$$W_\theta := \sum_{j=0}^{n-1} Z_j \mathbf{1}_{\{|Z_j| \leq \theta u\}}.$$

We first establish moment bounds for W_θ . Set $a_j := \gamma\rho^j$ and $M_j := \theta u/a_j$, so $Z_j = a_j \eta_{n-j} \stackrel{d}{=} a_j \eta_0$ and $\{|Z_j| > \theta u\} = \{|\eta_0| > M_j\}$. Since $M_j \geq \theta u/\gamma \rightarrow \infty$, Karamata's theorem [Resnick, 1987] applied to $|\eta_0|$ (whose tail is $\leq (1 + C_-)\bar{F}_v^+$ by (30)) gives

$$\mathbb{E}[|\eta_0| \mathbf{1}_{\{|\eta_0| > M_j\}}] \leq C M_j \bar{F}_v^+(M_j), \quad \mathbb{E}[\eta_0^2 \mathbf{1}_{\{|\eta_0| \leq M_j\}}] \leq C M_j^2 \bar{F}_v^+(M_j),$$

where the second bound uses $t < 2$. Summing the corresponding bounds for Z_j and using $\sum_j \bar{F}_v^+(M_j) = \Lambda_\gamma(\theta u)$,

$$\begin{aligned} \sum_j |\mathbb{E}[Z_j \mathbf{1}_{\{|Z_j| > \theta u\}}]| &\leq \sum_j a_j \mathbb{E}[|\eta_0| \mathbf{1}_{\{|\eta_0| > M_j\}}] \leq C \theta u \Lambda_\gamma(\theta u), \\ \sum_j \mathbb{E}[Z_j^2 \mathbf{1}_{\{|Z_j| \leq \theta u\}}] &\leq \sum_j a_j^2 \mathbb{E}[\eta_0^2 \mathbf{1}_{\{|\eta_0| \leq M_j\}}] \leq C (\theta u)^2 \Lambda_\gamma(\theta u). \end{aligned}$$

Since $\mathbb{E}Z_j = 0$, $\mathbb{E}W_\theta = -\sum_j \mathbb{E}[Z_j \mathbf{1}_{\{|Z_j| > \theta u\}}]$ and $\text{Var}(W_\theta) \leq \sum_j \mathbb{E}[Z_j^2 \mathbf{1}_{\{|Z_j| \leq \theta u\}}]$. Combining these with the shift stability $\Lambda_\gamma(\theta u) \sim \theta^{-t} \Lambda_\gamma(u)$ from (67), we conclude

$$|\mathbb{E}W_\theta| \leq C \theta^{1-t} u \Lambda_\gamma(u) = o(u), \quad \text{Var}(W_\theta) \leq C \theta^{2-t} u^2 \Lambda_\gamma(u), \quad (68)$$

where the $o(u)$ uses that $\Lambda_\gamma(u) \rightarrow 0$ and θ is fixed.

For the upper bound, partition the summands according to $\mathcal{L} := \{j : Z_j > \theta u\}$ and observe that on $\{S_n^{(v)} > u\}$ at least one of the following holds: some Z_j exceeds $(1 - \delta)u$; $|\mathcal{L}| \geq 2$; or $W_\theta > \delta u$ (since negative summands only reduce $S_n^{(v)}$). Therefore

$$\mathbb{P}(S_n^{(v)} > u) \leq \Lambda_\gamma((1 - \delta)u) + \mathbb{P}(|\mathcal{L}| \geq 2) + \mathbb{P}(W_\theta > \delta u). \quad (69)$$

By independence and (67), $\mathbb{P}(|\mathcal{L}| \geq 2) \leq \frac{1}{2}\Lambda_\gamma(\theta u)^2 \sim \frac{1}{2}\theta^{-2t}\Lambda_\gamma(u)^2 = o(\Lambda_\gamma(u))$. For the truncated sum, $|\mathbb{E}W_\theta| \leq \delta u/2$ for γ small enough by (68), and each centered summand $Z_j \mathbf{1}_{\{|Z_j| \leq \theta u\}} - \mathbb{E}[\cdot]$ is bounded by $2\theta u$. Bennett–Prokhorov’s inequality [Prokhorov, 1959] therefore gives

$$\mathbb{P}(W_\theta > \delta u) \leq \mathbb{P}(W_\theta - \mathbb{E}W_\theta > \delta u/2) \leq \exp\left\{-\frac{\delta}{8\theta} \operatorname{arsinh}\left(\frac{\delta \theta^{t-1}}{2C \Lambda_\gamma(u)}\right)\right\}.$$

Since $\operatorname{arsinh}(y) \sim \log(2y)$ as $y \rightarrow \infty$ and $\Lambda_\gamma(u) \rightarrow 0$, this bound equals $\Lambda_\gamma(u)^{\delta/(8\theta)+o(1)}$. The constraint $\theta < \delta/16$ forces the exponent to exceed 1, so

$$\mathbb{P}(W_\theta > \delta u) = o(\Lambda_\gamma(u)). \quad (70)$$

Substituting (70) and (67) into (69),

$$\limsup_{\gamma \downarrow 0} \frac{\mathbb{P}(S_n^{(v)} > u)}{\Lambda_\gamma(u)} \leq (1 - \delta)^{-t}.$$

For the lower bound, let $A_j := \{Z_j > (1 + \delta)u\}$ and $R_j := \sum_{i \neq j} Z_i$, which is independent of Z_j . On $A_j \cap \{R_j > -\delta u\}$ the full sum $S_n^{(v)} = Z_j + R_j$ exceeds u , so by Bonferroni,

$$\mathbb{P}(S_n^{(v)} > u) \geq \sum_{j=0}^{n-1} \mathbb{P}(A_j) \mathbb{P}(R_j > -\delta u) - \sum_{i < j} \mathbb{P}(A_i) \mathbb{P}(A_j),$$

and the double sum is bounded by $\frac{1}{2}\Lambda_\gamma((1 + \delta)u)^2 = o(\Lambda_\gamma(u))$ by independence and (67).

It remains to show $\sup_{0 \leq j < n} \mathbb{P}(R_j \leq -\delta u) \rightarrow 0$. Decompose the event $\{R_j \leq -\delta u\}$ according to whether some summand Z_i with $i \neq j$ is itself less than $-\theta u$. The probability of at least one such large negative summand is, by the union bound and the comparable left-tail condition (30),

$$\mathbb{P}(\exists i \neq j : Z_i < -\theta u) \leq \sum_{i \neq j} \mathbb{P}(Z_i < -\theta u) \leq C_- \Lambda_\gamma(\theta u) = O(\theta^{-t} \Lambda_\gamma(u))$$

uniformly in j , which is $o(1)$ for fixed θ since $\Lambda_\gamma(u) \rightarrow 0$. On the complementary event $\{Z_i \geq -\theta u \text{ for all } i \neq j\}$, the sum R_j coincides with the truncated sum

$$\widetilde{W}_\theta^{(j)} := \sum_{i \neq j} Z_i \mathbf{1}_{\{|Z_i| \leq \theta u\}} + \sum_{i \neq j} Z_i \mathbf{1}_{\{Z_i > \theta u\}},$$

whose first piece is a bounded centered sum (after centering) and whose second piece is non-negative. Hence on this event $R_j \geq \sum_{i \neq j} Z_i \mathbf{1}_{\{|Z_i| \leq \theta u\}} =: W_\theta^{(j)}$, and

$$\mathbb{P}(R_j \leq -\delta u) \leq \mathbb{P}(\exists i \neq j : Z_i < -\theta u) + \mathbb{P}(W_\theta^{(j)} \leq -\delta u).$$

The stronger absolute-mean estimate

$$\sum_{i=0}^{n-1} |\mathbb{E}[Z_i \mathbf{1}_{\{|Z_i| \leq \theta u\}}]| \leq C \theta^{1-t} u \Lambda_\gamma(u)$$

implies, uniformly in j ,

$$|\mathbb{E}W_\theta^{(j)}| \leq C \theta^{1-t} u \Lambda_\gamma(u) = o(u).$$

Moreover, dropping one index can only decrease the variance bound:

$$\operatorname{Var}(W_\theta^{(j)}) \leq C \theta^{2-t} u^2 \Lambda_\gamma(u).$$

Bennett–Prokhorov’s inequality applied to $\mathbb{E}W_\theta^{(j)} - W_\theta^{(j)}$ then gives, exactly as in the upper-bound argument,

$$\mathbb{P}(W_\theta^{(j)} \leq -\delta u) \leq \mathbb{P}(\mathbb{E}W_\theta^{(j)} - W_\theta^{(j)} > \delta u/2) = o(\Lambda_\gamma(u)) = o(1),$$

uniformly in j , for $\theta < \delta/16$. Combining these two estimates,

$$\sup_{0 \leq j < n} \mathbb{P}(R_j \leq -\delta u) \rightarrow 0,$$

and therefore

$$\mathbb{P}(S_n^{(v)} > u) \geq (1 - o(1))\Lambda_\gamma((1 + \delta)u) - o(\Lambda_\gamma(u)) \sim (1 + \delta)^{-t}\Lambda_\gamma(u),$$

giving $\liminf_{\gamma \downarrow 0} \mathbb{P}(S_n^{(v)} > u)/\Lambda_\gamma(u) \geq (1 + \delta)^{-t}$. Letting $\delta \downarrow 0$ proves $\mathbb{P}(S_n^{(v)} > u) \sim \Lambda_\gamma(u)$, and combination with (64) gives

$$\mathbb{P}(S_n^{(v)} > u) \sim \frac{1}{t\lambda} \frac{\bar{F}_v^+(u/\gamma)}{\gamma}.$$

It remains to transfer this to $y_n^{(v)} = S_n^{(v)} + \rho^n y_0^{(v)}$. The last condition in (31) gives $\rho^n |y_0^{(v)}| = o(u)$, so for any fixed $\epsilon \in (0, 1)$, eventually

$$\mathbb{P}(S_n^{(v)} > (1 + \epsilon)u) \leq \mathbb{P}(y_n^{(v)} > u) \leq \mathbb{P}(S_n^{(v)} > (1 - \epsilon)u).$$

Applying the asymptotic just established to $(1 \pm \epsilon)u$ together with shift stability (67), and then letting $\epsilon \downarrow 0$, gives (32); the second equivalence follows from $\bar{F}_v^+(u/\gamma) \sim h_v(u/\gamma) \gamma^t u^{-t}$. $\square$

C Proof of the Nonlinear Transfer

We first derive consequences of the primitive assumptions.

Lemma C.1 (Consequences of local and global conditions). *Assume Assumptions 2.1, 2.2, and 2.3. For every $q \in (1, p]$, there exists $C_q < \infty$ such that*

$$|m(x)| \leq C(1 + |x - x^*|), \quad (71)$$

$$\|m(x) - g(x, \varepsilon) + g(x^*, \varepsilon)\|_q \leq C_q |x - x^*|, \quad (72)$$

$$|H(x - x^*) - m(x)| \leq C \left(|x - x^*|^2 \mathbf{1}_{\{|x - x^*| \leq \kappa\}} + |x - x^*| \mathbf{1}_{\{|x - x^*| > \kappa\}} \right). \quad (73)$$

Moreover, for all sufficiently small γ ,

$$\|I_d - \gamma H\|_{\text{op}} \leq 1 - c_H \gamma \quad (74)$$

for some $c_H > 0$.

Proof. By Jensen and (6),

$$|m(x)| \leq \|g(x, \varepsilon)\|_1 \leq \|g(x, \varepsilon)\|_p \leq L_p(1 + |x - x^*|),$$

which gives (71). Let $D(x, \varepsilon) := m(x) - g(x, \varepsilon) + g(x^*, \varepsilon)$. If $|x - x^*| \leq \kappa$, then (7) and Jensen give $|m(x)| \leq L_g |x - x^*|$, and hence $\|D(x, \varepsilon)\|_q \leq C |x - x^*|$. If $|x - x^*| > \kappa$, then (71) and (6) imply the same bound after increasing the constant. This proves (72). The bound (73) follows from (9) on $\{|x - x^*| \leq \kappa\}$ and from linear growth on the complement.

Finally, (74) follows from Lemma B.1. $\square$

Lemma C.2 (Deterministic remainder bound). *Assume Assumptions 2.1, 2.2, and 2.3. Let $q \in ((p + 1)/2, p)$. Then there exist constants $C, c > 0$ such that*

$$\mathbb{E} |H(x_{n-1} - x^*) - m(x_{n-1})|^q \leq C\gamma^{p-1} + Ce^{-cn\gamma}(1 + |x_0 - x^*|^p). \quad (75)$$

Proof. By (73),

$$\begin{aligned} |H(x_{n-1} - x^*) - m(x_{n-1})|^q &\leq C |x_{n-1} - x^*|^{2q} \mathbf{1}_{\{|x_{n-1} - x^*| \leq \kappa\}} \\ &\quad + C |x_{n-1} - x^*|^q \mathbf{1}_{\{|x_{n-1} - x^*| > \kappa\}}. \end{aligned} \quad (76)$$

By (11),

$$\mathbb{P}(|x_{n-1} - x^*| > u) \leq C\gamma^{p-1} u^{-p} + Ce^{-c(n-1)\gamma} |x_0 - x^*|^p u^{-p}. \quad (77)$$

For the local part, the layer-cake identity gives

$$\mathbb{E}[|x_{n-1} - x^*|^{2q} \mathbf{1}_{\{|x_{n-1} - x^*| \leq \kappa\}}] \leq 2q \int_0^\kappa u^{2q-1} \mathbb{P}(|x_{n-1} - x^*| > u) du.$$

Split the integral at $\sqrt{\gamma}$. The part over $(0, \sqrt{\gamma})$ is $O(\gamma^q) = o(\gamma^{p-1})$, since $q > p - 1$. On $(\sqrt{\gamma}, \kappa)$, use (77); because $2q > p$, the integral is $O(\gamma^{p-1} + e^{-cn\gamma}|x_0 - x^*|^p)$. For the far part, since $q < p$,

$$\mathbb{E}[|x_{n-1} - x^*|^q \mathbf{1}_{\{|x_{n-1} - x^*| > \kappa\}}] \leq C(\gamma^{p-1} + e^{-cn\gamma}|x_0 - x^*|^p) \int_{\kappa}^{\infty} u^{q-p-1} du.$$

Combining the bounds proves (75). $\square$

Lemma C.3 (von Bahr–Esseen inequality for martingale differences). *Let $(d_k, \mathcal{F}_k)_{k=1}^n$ be $\mathbb{R}^d$ -valued martingale differences with $d_k \in L^q$ for some $1 < q \leq 2$. Then there exists $C_q < \infty$ such that*

$$\left\| \sum_{k=1}^n d_k \right\|_q^q \leq C_q \sum_{k=1}^n \|d_k\|_q^q. \quad (78)$$

Proof of Theorem 4.1. Let

$$R_k := H(x_{k-1} - x^*) - m(x_{k-1}), \quad D_k := m(x_{k-1}) - g(x_{k-1}, \varepsilon_k) + g(x^*, \varepsilon_k). \quad (79)$$

A direct subtraction of (2) and (14) gives

$$\Delta_n = (I_d - \gamma H)\Delta_{n-1} + \gamma R_n + \gamma D_n. \quad (80)$$

Iterating,

$$\Delta_n = I_{1,n} + I_{2,n}, \quad (81)$$

where

$$I_{1,n} := \gamma \sum_{k=1}^n (I_d - \gamma H)^{n-k} R_k, \quad I_{2,n} := \gamma \sum_{k=1}^n (I_d - \gamma H)^{n-k} D_k. \quad (82)$$

By (74), $\|I_d - \gamma H\|_{\text{op}} \leq 1 - c_H \gamma$.

For $I_{1,n}$, Minkowski's inequality and Lemma C.2 imply

$$\|I_{1,n}\|_q \leq C\gamma \sum_{k=1}^n (1 - c_H \gamma)^{n-k} \left(\gamma^{(p-1)/q} + e^{-ck\gamma/q} (1 + |x_0 - x^*|^{p/q}) \right).$$

The geometric convolution gives

$$\|I_{1,n}\|_q^q \leq C\gamma^{p-1} + Ce^{-cn\gamma}(1 + |x_0 - x^*|^p). \quad (83)$$

For $I_{2,n}$, $(D_k, \mathcal{F}_k)$ is a martingale-difference sequence. Thus $d_{n,k} := \gamma(I_d - \gamma H)^{n-k} D_k$ is a martingale-difference array. By Lemma C.3,

$$\|I_{2,n}\|_q^q \leq C\gamma^q \sum_{k=1}^n (1 - c_H \gamma)^{q(n-k)} \|D_k\|_q^q.$$

By (72) and Theorem 2.5 with exponent q ,

$$\|D_k\|_q^q \leq C(1 - c_q \gamma)^{k-1} |x_0 - x^*|^q + C\gamma^{q-1}.$$

Substitution gives

$$\|I_{2,n}\|_q^q \leq C\gamma^{2q-2} + Ce^{-cn\gamma}(1 + |x_0 - x^*|^p). \quad (84)$$

Since $q > (p+1)/2$, we have $2q-2 > p-1$, so $\gamma^{2q-2} = o(\gamma^{p-1})$. Combining (81), (83), and (84) proves (38). $\square$

D Proof of the Nonlinear Tail Transfer

We first prove a general-threshold version. The power-threshold theorem in the main text follows by specializing to $u_\gamma = \gamma^a$.

Lemma D.1 (Small-shift sandwich). *For every $\ell \in \mathbb{R}^d \setminus \{0\}$, $u > 0$, and $c \in (0, 1)$,*

$$\mathbb{P}(y_n^{(\ell)} > (1+c)u) - \mathbb{P}(|\ell^\top \Delta_n| > cu) \leq \mathbb{P}(x_n^{(\ell)} > u) \leq \mathbb{P}(y_n^{(\ell)} > (1-c)u) + \mathbb{P}(|\ell^\top \Delta_n| > cu).$$

Proof. Since $x_n^{(\ell)} = y_n^{(\ell)} + \ell^\top \Delta_n$,

$$\{y_n^{(\ell)} > (1+c)u, |\ell^\top \Delta_n| \leq cu\} \subseteq \{x_n^{(\ell)} > u\},$$

and

$$\{x_n^{(\ell)} > u, |\ell^\top \Delta_n| \leq cu\} \subseteq \{y_n^{(\ell)} > (1-c)u\}.$$

The claim follows immediately. $\square$

Proposition D.2 (General-threshold nonlinear transfer). *Assume Assumptions 2.1, 2.2, 2.3, and 3.1. Let $\ell \in \mathbb{R}^d \setminus \{0\}$ and assume*

$$C_\ell := \int_0^\infty \mu_\eta\{z : \ell^\top e^{-sH} z > 1\} ds > 0.$$

Let $u_\gamma \downarrow 0$ and $n_\gamma \rightarrow \infty$ satisfy Assumption 3.2. Suppose that

$$\frac{n_\gamma \gamma}{\log(1/\gamma)} \rightarrow \infty$$

and that, for some $q \in ((p+1)/2, p)$,

$$\frac{\gamma^{p-1}}{u_\gamma^q} = o\left(\frac{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)}{\gamma}\right).$$

Then

$$\mathbb{P}(x_{n_\gamma}^{(\ell)} > u_\gamma) \sim \mathbb{P}(y_{n_\gamma}^{(\ell)} > u_\gamma) \sim C_\ell \frac{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)}{\gamma}.$$

Proof. Fix $c \in (0, 1)$. By Corollary 3.5, applied to $(1 \pm c)u_\gamma$,

$$\mathbb{P}(y_{n_\gamma}^{(\ell)} > (1 \pm c)u_\gamma) \sim C_\ell \frac{\mathbb{P}(|\eta_0| > (1 \pm c)u_\gamma/\gamma)}{\gamma}.$$

Regular variation gives

$$\frac{\mathbb{P}(|\eta_0| > (1 \pm c)u_\gamma/\gamma)}{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)} \rightarrow (1 \pm c)^{-t}.$$

Hence

$$\mathbb{P}(y_{n_\gamma}^{(\ell)} > (1 \pm c)u_\gamma) \sim (1 \pm c)^{-t} C_\ell \frac{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)}{\gamma}.$$

It remains to show that the nonlinear approximation error is negligible. By Markov's inequality and Theorem 4.1,

$$\begin{aligned} \mathbb{P}(|\ell^\top \Delta_{n_\gamma}| > cu_\gamma) &\leq \frac{|\ell|^q \|\Delta_{n_\gamma}\|_q^q}{c^q u_\gamma^q} \\ &\leq C_c u_\gamma^{-q} \{\gamma^{p-1} + e^{-c_q n_\gamma \gamma} (1 + |x_0 - x^*|^p)\}. \end{aligned}$$

The first term is

$$o\left(\frac{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)}{\gamma}\right)$$

by assumption. For the initialization term, Potter's bound [Resnick, 1987] gives, for every small $\varepsilon > 0$ and all sufficiently small γ ,

$$\frac{1}{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)} \leq C \left(\frac{u_\gamma}{\gamma}\right)^{t+\varepsilon}.$$

Since $q < t$, choose $\varepsilon > 0$ so small that $q < t + \varepsilon$. Then

$$\begin{aligned} &\frac{u_\gamma^{-q} e^{-c_q n_\gamma \gamma}}{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)/\gamma} \\ &\leq C e^{-c_q n_\gamma \gamma} \gamma u_\gamma^{-q} \left(\frac{u_\gamma}{\gamma}\right)^{t+\varepsilon} \\ &= C e^{-c_q n_\gamma \gamma} \gamma^{1-t-\varepsilon} u_\gamma^{t+\varepsilon-q} \rightarrow 0, \end{aligned}$$

because $u_\gamma \downarrow 0$, $t + \varepsilon - q > 0$, and $n_\gamma \gamma / \log(1/\gamma) \rightarrow \infty$. Therefore

$$\mathbb{P}(|\ell^\top \Delta_{n_\gamma}| > cu_\gamma) = o\left(\frac{\mathbb{P}(|\eta_0| > u_\gamma/\gamma)}{\gamma}\right).$$

Applying Lemma D.1, dividing by $C_\ell \mathbb{P}(|\eta_0| > u_\gamma/\gamma)/\gamma$, and using the preceding estimates gives

$$(1+c)^{-t} \leq \liminf_{\gamma \downarrow 0} \frac{\mathbb{P}(x_{n_\gamma}^{(\ell)} > u_\gamma)}{C_\ell \mathbb{P}(|\eta_0| > u_\gamma/\gamma)/\gamma}$$

and

$$\limsup_{\gamma \downarrow 0} \frac{\mathbb{P}(x_{n_\gamma}^{(\ell)} > u_\gamma)}{C_\ell \mathbb{P}(|\eta_0| > u_\gamma/\gamma)/\gamma} \leq (1-c)^{-t}.$$

Letting $c \downarrow 0$ proves the claim. $\square$

Proof of Theorem 4.2. Apply Proposition D.2 with $u_\gamma = \gamma^a$. We only need to verify its general-threshold assumptions.

First, $u_\gamma \downarrow 0$ and $u_\gamma/\gamma = \gamma^{a-1} \rightarrow \infty$ because $0 < a < 1$. Since

$$0 < a < \frac{t-1}{t},$$

Potter's bound [Resnick, 1987] implies

$$\frac{\mathbb{P}(|\eta_0| > \gamma^{a-1})}{\gamma} \rightarrow 0.$$

The burn-in condition

$$\frac{n_\gamma \gamma}{\log(1/\gamma)} \rightarrow \infty$$

implies both $n_\gamma \gamma \rightarrow \infty$ and

$$\frac{e^{-\alpha n_\gamma \gamma / 2}}{\gamma^a} \rightarrow 0.$$

Thus Assumption 3.2 holds.

It remains to verify the nonlinear-transfer condition. By Potter's bound [Resnick, 1987], for every sufficiently small $\varepsilon > 0$,

$$\mathbb{P}(|\eta_0| > \gamma^{a-1}) \geq C_\varepsilon^{-1} \gamma^{(1-a)(t+\varepsilon)}$$

for all sufficiently small γ . Hence

$$\frac{\gamma^{p-1-aq}}{\mathbb{P}(|\eta_0| > \gamma^{a-1})/\gamma} \leq C_\varepsilon \gamma^{p-t+a(t-q)-\varepsilon(1-a)}.$$

Since $p - t + a(t - q) > 0$, choosing $\varepsilon > 0$ small enough gives

$$\gamma^{p-1-aq} = o\left(\frac{\mathbb{P}(|\eta_0| > \gamma^{a-1})}{\gamma}\right).$$

Proposition D.2 now yields the result. $\square$

Proof of Corollary 4.3. Let $u_\gamma = \gamma^a$. The assumptions

$$0 < a < \frac{t-1}{t}, \quad \frac{n_\gamma \gamma}{\log(1/\gamma)} \rightarrow \infty$$

imply the threshold and burn-in conditions of Theorem 3.7. Therefore

$$\mathbb{P}(y_{n_\gamma}^{(v)} > \gamma^a) \sim \frac{1}{t\lambda_v} \frac{\bar{F}_v^+(\gamma^{a-1})}{\gamma} \sim \frac{1}{t\lambda_v} h_v(\gamma^{a-1}) \gamma^{t-1-at}.$$

Moreover, for each fixed $c \in (0, 1)$,

$$\mathbb{P}(y_{n_\gamma}^{(v)} > (1 \pm c)\gamma^a) \sim (1 \pm c)^{-t} \mathbb{P}(y_{n_\gamma}^{(v)} > \gamma^a).$$

By Markov's inequality and Theorem 4.1,

$$\mathbb{P}(|\Delta_{n_\gamma}^{(v)}| > c\gamma^a) \leq C_c \gamma^{-aq} \left\{ \gamma^{p-1} + e^{-c_q n_\gamma \gamma} (1 + |x_0 - x^*|^p) \right\}.$$

The stationary part is negligible relative to $h_v(\gamma^{a-1})\gamma^{t-1-at}$ because

$$\frac{\gamma^{p-1-aq}}{h_v(\gamma^{a-1})\gamma^{t-1-at}} = \frac{\gamma^{p-t+a(t-q)}}{h_v(\gamma^{a-1})} \rightarrow 0$$

by Potter's bound [Resnick, 1987] and the condition $p - t + a(t - q) > 0$. The initialization part is also negligible because $n_\gamma \gamma / \log(1/\gamma) \rightarrow \infty$ makes $e^{-c_q n_\gamma \gamma}$ smaller than any polynomial power of γ .

Thus

$$\mathbb{P}(|\Delta_{n_\gamma}^{(v)}| > c\gamma^a) = o\left(\mathbb{P}(y_{n_\gamma}^{(v)} > \gamma^a)\right).$$

Applying Lemma D.1 with $\ell = v$, dividing by $\mathbb{P}(y_{n_\gamma}^{(v)} > \gamma^a)$, and then letting $c \downarrow 0$ gives

$$\mathbb{P}(x_{n_\gamma}^{(v)} > \gamma^a) \sim \mathbb{P}(y_{n_\gamma}^{(v)} > \gamma^a).$$

Combining this with the linear tail asymptotic proves the corollary. □

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